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DoorCo AI QC System

Door Skin Defect Detection

- 📊 Executive Dashboard
- 🏭 Production Monitor
- 🔬 Panel Inspector
- 📈 Model Performance
- 📚 Defect Library

ACTIVE MODEL

🏆 **GradientBoosting**

Accuracy	**97.9%**
F1 Weighted	**0.9785**
F1 Macro	**0.9808**
Defect Classes	**8 types**
Lines Deployed	**Multi-site**

📚 Door Skin Defect Library

Visual reference · 8 defect types · Severity classifications · Remediation actions

GOOD None

$0/escape

Meets all DoorCo quality standards. No defects detected.

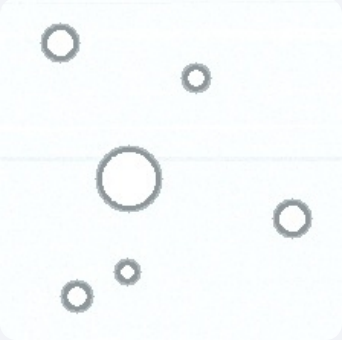
Action: Clear for shipment

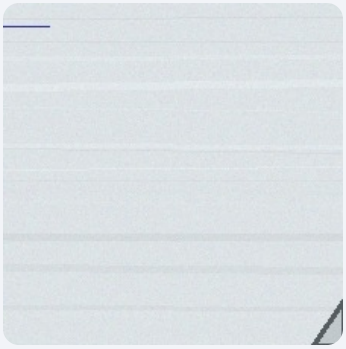
CRACK Major

$180/escape

Structural crack in door skin. Door fails structural integrity test.

Action: Scrap — structural failure





BLISTER

Moderate

$85/escape

Sub-surface air bubble causing paint/skin to lift.

Action: Sand and refinish or reject

CROOKED CORNER

Major

$120/escape

Corner geometry outside tolerance. Door will not fit frame.

Action: Scrap — dimensional reject

THIN PAINT

Minor

$35/escape

Insufficient paint thickness. Fails warranty and weather resistance spec.

Action: Re-coat on paint line

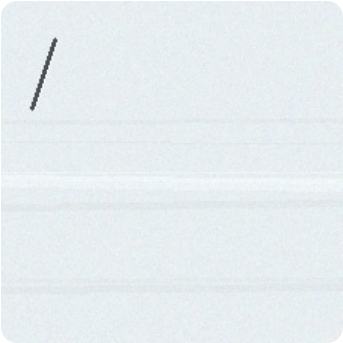
THICK PAINT

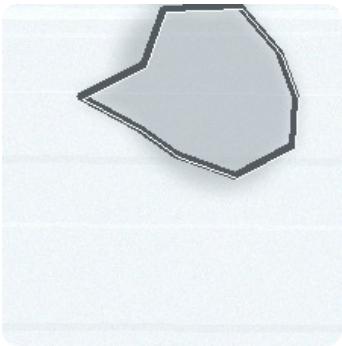
Minor

$35/escape

Excessive paint buildup. Causes door binding and appearance defect.

Action: Sand down or re-coat





SCRATCH

Moderate

$55/escape

Surface scratch from handling, conveyor, or tooling contact.

Action: Touch-up or reject based on depth

DELAMINATION

Major

$200/escape

Door skin separating from substrate. Structural failure.

Action: Immediate scrap — safety reject